

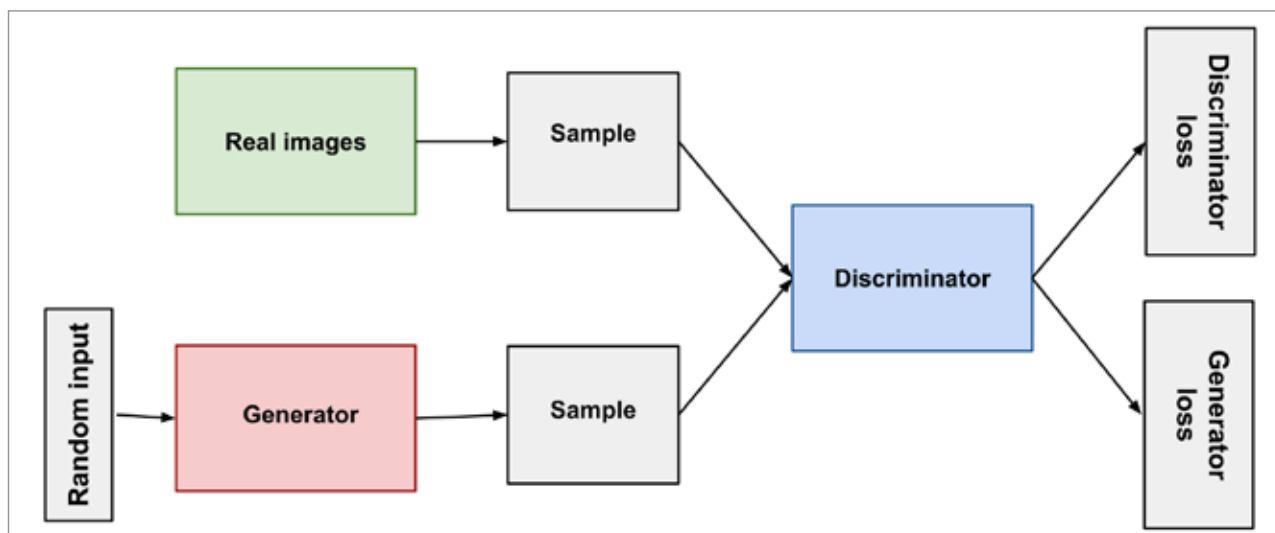


Figure 1: Architecture of GAN

networks are trained together in an adversarial setup, forming a minimax optimisation problem in which G tries to fool D and D learns to detect fakes.

Figure 1 illustrates this interaction, which can be viewed as a two-player game. As training progresses, the generator becomes better at producing realistic data while the discriminator sharpens its detection ability. Successful training allows GANs to approximate the true data distribution, leading to highly realistic

outputs. Models such as StyleGAN demonstrated this capability through convincing face and scene synthesis.

GANs are especially known for producing sharp, detailed images, making them effective for tasks like image synthesis and super-resolution. However, they are difficult to train and prone to issues such as mode collapse or unstable convergence. They also do not provide explicit likelihood estimates. Despite these drawbacks, GANs remain widely used in computer vision due to

their strong visual quality.

Variational Autoencoders (VAEs):

Variational Autoencoders (VAEs) take a probabilistic approach to generative modelling. Introduced by Kingma and Welling in 2013, VAEs extend the traditional autoencoder by representing each input as a distribution in latent space rather than a fixed vector. The encoder outputs parameters of this distribution, usually a Gaussian, while the decoder samples from it to reconstruct data. Training optimises

Model type	Year	Key idea	Strengths	Challenges
Generative Adversarial Network (GAN)	2014 (Goodfellow <i>et al</i>)	Uses two neural networks (generator and discriminator) trained in a minimax adversarial setup	Generates high-fidelity, sharp samples; very fast inference	Difficult to train due to instability; suffers from mode collapse; lacks explicit likelihood function
Variational Autoencoder (VAE)	2013 (Kingma & Welling)	Learns probabilistic latent encodings using KL-divergence regularisation	Stable training process; interpretable latent space; enables likelihood estimation	Outputs can appear blurry; requires careful balance between reconstruction loss and regularisation
Diffusion model	2015/2020 (Sohl-Dickstein <i>et al</i> ; Ho <i>et al</i>)	Recovers data from noise by reversing a diffusion process through iterative denoising	Achieves state-of-the-art generation quality; avoids mode collapse	Sampling is slow; computationally expensive; requires many denoising steps
Transformer-based models (e.g., GPT)	2017/2020 (Vaswani <i>et al</i> ; Brown <i>et al</i>)	Uses self-attention mechanisms to learn long-range dependencies in an autoregressive manner	Highly effective for language tasks; processes long contexts; flexible across domains	Requires significant computational resources; may hallucinate facts; interpretability remains difficult

Table 1: Comparison of major generative model types in deep learning